- Vectors in underline, matrices in capital. Assume all matrices have real entries unless stated otherwise.
- For a vector $\underline{x}$, we denote by x_i the i^{th} entry of $\underline{x}$.
- The Hessian of the least-squares objective $f(\underline{x}) = \frac{1}{2n}\left\|A\underline{x} - \underline{y}\right\|^2$ is $H = \frac{1}{n}A^\top A$ with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_d > 0$. The condition number is $\kappa = \lambda_1/\lambda_d$.
- Use the companion notebook `AI2113-A5_GD.ipynb` for all coding parts. Complete the skeleton functions marked with `TODO`.

1. *Gradient Descent: Convergence to ε-Accuracy.*

   Consider the least-squares problem $\min_{\underline{x}} f(\underline{x}) = \frac{1}{2n}\left\|A\underline{x} - \underline{y}\right\|^2$ with $A \in \mathbb{R}^{n\times d}$, $n \geq d$, and A having full column rank. Run GD with step size $\eta = 1/\lambda_1$ starting from $\underline{x}^{(0)} = \underline{0}$.

   (a) *(Coding.)* Using the companion notebook, generate least-squares problems with controlled condition numbers $\kappa \in \{2, 5, 10, 50, 100\}$. For each κ and each $\varepsilon \in \{10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}\}$:

      (i) Run GD (with $\eta = 1/\lambda_1$) and record the number of iterations k_{sim} needed to first achieve $f(\underline{x}^{(k)}) - f(\underline{x}^*) \leq \varepsilon$.

      (ii) Compute the theoretical upper bound k_{theory} from part (a).

      (iii) Report both values in a table. How tight is the bound? When is the gap largest?

   (b) *(Theory.)* Now consider the optimal step size $\eta^* = 2/(\lambda_1 + \lambda_d)$. Show that the function-value contraction becomes $\left(\frac{\kappa-1}{\kappa+1}\right)^2$, and derive the corresponding iteration bound. How does this compare quantitatively with the bound from part (a) for large κ?

2. *Early Stopping as Implicit Regularization.*

   Consider the statistical model $\underline{y} = A\underline{x}^* + \underline{\varepsilon}$, $\underline{\varepsilon} \sim \mathcal{N}(0, \sigma^2 I_n)$. Run GD from $\underline{x}^{(0)} = \underline{0}$ on the training objective.

   (a) *(Coding.)* Fix $n = 80$, $d = 70$, $\sigma = 3$. Using the notebook:

      (i) Run GD for $K = 500$ iterations and record the test error $R(k) = \frac{1}{n_{\text{test}}}\left\|A_{\text{test}}\underline{x}^{(k)} - \underline{y}_{\text{test}}\right\|^2$ at each iteration.

      (ii) Average over $T = 50$ independent trials (fresh data each trial).

      (iii) Plot the averaged test error $\bar{R}(k)$ and identify the optimal stopping iteration k^*.

   (b) *(Coding.)* On the same plot, add the best ridge regression test error (optimizing λ over a grid). Which method achieves lower test error: early-stopped GD or optimally-tuned ridge? Explain why, in terms of the *filter factors* applied to each eigencomponent:

   $$\text{GD (k steps): } 1 - (1 - \eta\lambda_i)^k, \qquad \text{Ridge (λ): } \frac{\lambda_i}{\lambda_i + \lambda}.$$

3. *Stochastic Gradient Descent: Learning Rate Schedules.*

   Consider SGD for least squares: at step k, pick i_k uniformly at random from $\{1, \ldots, n\}$ and update

   $$\underline{x}^{(k+1)} = \underline{x}^{(k)} - \eta_k \left(\underline{a}_{i_k}^\top \underline{x}^{(k)} - y_{i_k}\right)\underline{a}_{i_k},$$

   where $\underline{a}_i^\top$ is the i-th row of A.

   (a) *(Coding.)* Using the notebook, implement SGD and run it on a noiseless least-squares problem ($n = 200$, $d = 30$) with each of the following step-size schedules. For each schedule, plot $\mathbb{E}\left\|\underline{x}^{(k)} - \underline{x}^*\right\|^2$ (averaged over 50 runs) vs. iteration k on a log-log plot.

      (i) Fixed step size: $\eta_k = \eta$ (small constant).

(ii) $\eta_k = c/k$ with $c > 1/(2\lambda_d)$ (correct constant).

(iii) $\eta_k = c'/k$ with $c' < 1/(2\lambda_d)$ (incorrect/too-small constant).

(iv) $\eta_k = c/k^2$.

(v) $\eta_k = c/\sqrt{k}$.

Describe the qualitative behavior you observe for each schedule. Which converge to the optimum? At what rate?

(b) *(Theory.)* Consider the schedule $\eta_t = c/t^2$. Write down the corresponding ODE for $E(t)$ and solve it. Show that the error converges, but at a rate *slower* than $O(1/t)$. Why does shrinking the step size too aggressively hurt convergence?

(c) *(Theory.)* Repeat for $\eta_t = c/\sqrt{t}$. Show that the steady-state error does not vanish (similar to fixed step size, but with a smaller residual). Does your theoretical prediction match the simulation from part (a)(v)?

4. *Strong Convexity and Smoothness.*

(a) For each function below, determine whether it is (i) μ-strongly convex, and (ii) L-smooth. If so, find the best (largest μ, smallest L) parameters and the condition number $\kappa = L/\mu$.

(i) $f(\underline{x}) = \frac{1}{2}\underline{x}^\top Q\,\underline{x} + \underline{b}^\top \underline{x}$, where $Q \succ 0$ is symmetric with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_d > 0$.

(ii) $f(x) = x^4 + x^2$, $x \in \mathbb{R}$.

(iii) $f(\underline{x}) = \log\big(\sum_{i=1}^n e^{\underline{a}_i^\top \underline{x}}\big)$, the log-sum-exp function.

5. *True or False.*

For each statement below, determine whether it is true or false. If true, provide a proof. If false, provide an explicit counterexample.

Gradient descent and smoothness.

(i) Let f be L-smooth (not necessarily convex) and *bounded below*, i.e. $f^* = \inf_{\underline{x}} f(\underline{x}) > -\infty$. If we run GD with $\eta = 1/L$ for T iterations, then

$$\min_{0 \leq k \leq T-1} \left\|\nabla f(\underline{x}^{(k)})\right\|^2 \;\leq\; \frac{2L\big(f(\underline{x}^{(0)}) - f^*\big)}{T}.$$

That is, GD must produce *at least one* iterate with a small gradient, even without convexity.

(ii) Is the "bounded below" assumption in the above part essential ? Consider $f(x) = -x^2$. Is this function $L-$ smooth, if so what is the smallest such L ? Show that GD with $\eta = 1/2$ starting from $x^{(0)} = 1$ produces iterates that diverge to ∞, and $|\nabla f(x^{(k)})|$ grows without bound.

(iii) Let $f : \mathbb{R}^d \to \mathbb{R}$ be L-smooth and *convex* (but not necessarily strongly convex). Then GD with $\eta = 1/L$ satisfies

$$f(\underline{x}^{(T)}) - f(\underline{x}^*) \;\leq\; \frac{L\left\|\underline{x}^{(0)} - \underline{x}^*\right\|^2}{2T}.$$

Strong convexity and conjugates.

(iv) Let $f : \mathbb{R}^d \to \mathbb{R}$ be μ-strongly convex and differentiable. Then for any $\underline{r} \in \mathbb{R}^d$, there is *at most one* $\underline{x}$ satisfying $\nabla f(\underline{x}) = \underline{r}$.

(v) If $f : \mathbb{R}^d \to \mathbb{R}$ is L-smooth, then f is convex.

(vi) Let $f : \mathbb{R}^d \to \mathbb{R}$ be μ-strongly convex and differentiable. Then f is L-smooth for some finite L.

(vii) If f is closed, convex, and L-smooth, then f^* (the Fenchel conjugate) is $\frac{1}{L}$-strongly convex.

(viii) If f is closed and μ-strongly convex, then f^* is μ-smooth.

(ix) Let $g : \mathbb{R}^d \to \mathbb{R}$ be convex (but not strongly convex). Then $h(\underline{x}) = g(\underline{x}) + \frac{\mu}{2} \|\underline{x}\|^2$ is μ-strongly convex.

Stochastic gradient descent.

(x) SGD for least squares is a *descent method*: at every step, $f(\underline{x}^{(k+1)}) \le f(\underline{x}^{(k)})$.

(xi) The SGD gradient estimate is unbiased: $\mathbb{E}[\underline{g}^{(k)} \mid \underline{x}^{(k)}] = \nabla f(\underline{x}^{(k)})$.

(xii) SGD with fixed step size η converges to the minimizer $\underline{x}^*$ of the least-squares objective, i.e. $\mathbb{E} \left\|\underline{x}^{(k)} - \underline{x}^*\right\|^2 \to 0$ as $k \to \infty$.

(xiii) For least-squares objectives, full GD converges to $\underline{x}^*$ from *any* initialization $\underline{x}^{(0)}$ (provided $\eta < 2/\lambda_1$).

(xiv) SGD with the schedule $\eta_k = c/k$ ($c > 1/(2\lambda_d)$) achieves linear convergence, i.e. $\mathbb{E} \left\|\underline{x}^{(k)} - \underline{x}^*\right\|^2 \le C \rho^k$ for some $\rho < 1$.

References